



Lift Station 6-Month PdM Study

Test plan · acceptance criteria · results

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Product	MooreVIEW MVP Suite (est-pc)
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1. Purpose

This document validates the **duplex lift station 6-month predictive maintenance (PdM) study** across four MCSA / inference scenarios. It records **test criteria**, **automated checks**, and **observed results** from the simulated restaurant duplex lift station (180 days, 8 PdM assets, CMMS auto work orders).

2. Scenario matrix (acceptance design)

ID	Title	MCSA source	Host path	Expected early warning
opta-mcsa-regression	Opta pseudo MCSA uplink — PdM regression only	Opta pseudo MCSA	None (PdM regression only)	Late (month 6+)
opta-mcsa-onnx	Opta pseudo MCSA — host ONNX	Opta pseudo MCSA	Host onnx	Early (month 1)
true-mcsa-regression	True MCSA (FFT sim) — host rules	True FFT MCSA (sim)	Host rule	Early (month 1)
true-mcsa-onnx	True MCSA (FFT sim) — host ONNX	True FFT MCSA (sim)	Host onnx	Early (month 1)

3. Automated unit tests

Run before or after the study:

```
cd est-pc
node --test test/liftStationStudy.test.js test/hostInference.test.js
```

Suite	Criterion	Result
liftStationStudy.test.js	Four scenario IDs match design matrix	PASS
liftStationStudy.test.js	Regression scenario produces zero edge docs in sim	PASS
liftStationStudy.test.js	Production early-warning profile (host supplement + CMMS)	PASS
hostInference.test.js	MCSA feature vector, registry, host orchestration	PASS
Summary	14 passed · 0 failed (exit 0)	PASS

3.1 Unit test definitions

Test	Pass criterion
Scenario matrix	STUDY_SCENARIOS = opta-mcsa-regression, opta-mcsa-onnx, true-mcsa-regression, true-mcsa-onnx with hostInference false / onnx / rule / onnx

Asset registration	8 study assets + live pump-1 / pump-2; all three PdM forecast methods enabled
Production profile	<code>inference.hostEnabled, mode: host-supplement, cmms.autoWorkOrdersFromPdm</code>
True MCSA sim	FFT channels include bearing and eccentricity metadata
Regression sim	<code>edgeDocs.length === 0</code> with >50 start events over 90-day sample
MCSA features	64-dim vector from Opta lite telemetry fixture
Host inference	Supplement skips when device <code>edgeAi</code> present; runs for HVAC without edge

4. Integration study criteria

#	Criterion	Target
I1	Simulated duration	180 days
I2	Registered PdM assets	8 (pump-{1,2}-{scenario})
I3	MCSA uplink reports per scenario	~7,400 (all motor starts)
I4	Pen historian samples per scenario	~45,120
I5	Total proactive CMMS work orders	8 (2 per scenario)
I6	WO source	pdm with urgent priority
I7	Regression-only path	Latest WO month (≥ 6) vs inference paths (≤ 1)
I8	PdM failure threshold	0.35 health index
I9	Forecast methods stacked	health_index, run_amps_creep, start_time_ms (shortest RUL wins)
I10	ONNX model	<code>data/models/lift-submersible-v3.onnx</code>

5. Study results by scenario

Opta pseudo MCSA uplink — PdM regression only (opta-mcsa-regression)

Metric	Value
MCSA uplink reports	7400
Host / edge inferences	0
Pen samples	45120
Proactive work orders	2
First alert month	6

WO	Asset	Priority	First alert month
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WO-0019	pump-2-opta-mcsa-regression	urgent	6
WO-0018	pump-1-opta-mcsa-regression	urgent	6

Acceptance check	Result
No host inference docs written (regression-only path)	PASS
MCSA uplink reports generated (lite mode)	PASS
Exactly 2 proactive CMMS work orders (lead + lag pump)	PASS
First proactive WO issued in month 6 or later (SCADA/regression lag)	PASS

Final forecasts (end of study):

Asset	Severity	Health	Headline
pump-1-opta-mcsa-regression	failed	—	pump-1-opta-mcsa-regression: At or past predicted run amp creep / impeller load — service now
pump-2-opta-mcsa-regression	failed	—	pump-2-opta-mcsa-regression: At or past predicted run amp creep / impeller load — service now

Opta pseudo MCSA — host ONNX (opta-mcsa-onnx)

Metric	Value
MCSA uplink reports	7400
Host / edge inferences	7400
Pen samples	45120
Proactive work orders	2
First alert month	1

WO	Asset	Priority	First alert month
WO-0021	pump-2-opta-mcsa-onnx	urgent	1
WO-0020	pump-1-opta-mcsa-onnx	urgent	1

Acceptance check	Result
Host inference runs on every motor start (onnx)	PASS
MCSA uplink reports generated (lite mode)	PASS
Exactly 2 proactive CMMS work orders (lead + lag pump)	PASS
First proactive WO issued by month 1 (inference early warning)	PASS

Final forecasts (end of study):

Asset	Severity	Health	Headline
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pump-1-opta-mcsa-onnx	failed	0.12	pump-1-opta-mcsa-onnx: At or past predicted health threshold breach — service now
pump-2-opta-mcsa-onnx	failed	0.12	pump-2-opta-mcsa-onnx: At or past predicted health threshold breach — service now

True MCSA (FFT sim) — host rules (true-mcsa-regression)

Metric	Value
MCSA uplink reports	7400
Host / edge inferences	7400
Pen samples	45120
Proactive work orders	2
First alert month	1

WO	Asset	Priority	First alert month
WO-0023	pump-2-true-mcsa-regression	urgent	1
WO-0022	pump-1-true-mcsa-regression	urgent	1

Acceptance check	Result
Host inference runs on every motor start (rule)	PASS
MCSA uplink reports generated (fft mode)	PASS
Exactly 2 proactive CMMS work orders (lead + lag pump)	PASS
First proactive WO issued by month 1 (inference early warning)	PASS

Final forecasts (end of study):

Asset	Severity	Health	Headline
pump-1-true-mcsa-regression	failed	0.12	pump-1-true-mcsa-regression: At or past predicted health threshold breach — service now
pump-2-true-mcsa-regression	failed	0.12	pump-2-true-mcsa-regression: At or past predicted health threshold breach — service now

True MCSA (FFT sim) — host ONNX (true-mcsa-onnx)

Metric	Value
MCSA uplink reports	7400
Host / edge inferences	7400
Pen samples	45120

Proactive work orders	2
First alert month	1

WO	Asset	Priority	First alert month
WO-0025	pump-2-true-mcsa-onnx	urgent	1
WO-0024	pump-1-true-mcsa-onnx	urgent	1

Acceptance check	Result
Host inference runs on every motor start (onnx)	PASS
MCSA uplink reports generated (fft mode)	PASS
Exactly 2 proactive CMMS work orders (lead + lag pump)	PASS
First proactive WO issued by month 1 (inference early warning)	PASS

Final forecasts (end of study):

Asset	Severity	Health	Headline
pump-1-true-mcsa-onnx	failed	0.12	pump-1-true-mcsa-onnx: At or past predicted health threshold breach — service now
pump-2-true-mcsa-onnx	failed	0.12	pump-2-true-mcsa-onnx: At or past predicted health threshold breach — service now

6. Cross-scenario comparison

Scenario	First WO month	Edge inferences	Forecast driver (regression path)
opta-mcsa-regression	6	0	run_amps_creep (no health index from inference)
opta-mcsa-onnx	1	7400	health_index (inference-fed)
true-mcsa-regression	1	7400	health_index (inference-fed)
true-mcsa-onnx	1	7400	health_index (inference-fed)

Early-warning delta (Opta pseudo MCSA): regression-only detected failure **5 simulated month(s)** later than host ONNX (6 vs 1).

7. Overall verdict

Gate	Result
Automated unit tests	PASS
All scenario acceptance checks (16)	PASS
Total proactive WOs = 8	PASS (8)
Release recommendation	PASS

8. Manual verification in MooreVIEW UI

After loading study settings:

1. **Historian** → **Logger config** → **PdM** — confirm 8 assets (`pump-1-*` / `pump-2-*`).
2. **CMMS** (`/cmms`) — filter **source: pdm** — verify WO-0018 through WO-0025 (or current run).
3. **Per-scenario PDFs** — open `PdM_{scenario}_pump-*.pdf` under each scenario folder in the study output directory.
4. **Production profile** — review `data/pdm-production-early-warning.json` ; apply with `npm run pdm:production-profile` .

9. Reproduce

```
cd est-pc
py -3 scripts/gen-lift-study-onnx.py           # if model missing
npm run study:ls-6mo                           # ~30 min, requires MongoDB
npm run build:ls-6mo-study-test-pdf            # this document
```

Study artifacts: C:/Users/Public/data/est-pc/data/sim-studies/ls-6mo-4way-2026-07-25T12-33-53

10. Sign-off

Role	Name	Date	Signature
Test engineer			
PdM / reliability			
Product owner			